

Practice Midterm

Form P · Practice

Practice for the midterm exam · not graded · use it to time yourself

Name: _____ Student ID: _____

40 points | Suggested working time 40 minutes

Use a basic calculator. Show your working. Allow about 7, 10, 7, 10 and 3 minutes for Sections 1–5, plus 3 minutes to read and check. Sections 1 and 5 are marked on clear reasoning, not on matching one “right” answer. The data are synthetic teaching data from a fictional seaside café. This practice trains the same skills as the real midterm through different questions and different results.

Section	What it covers	Points	Score
1	Choose the inputs	8	
2	Read the LINEST output	12	
3	Errors and outliers	8	
4	Did the price change sales?	9	
5	Summary for the owner	3	
Total		40	

SECTION 1 · Choose the inputs

8 points · about 7 minutes

A seaside café wants to predict **tomorrow’s pastry sales** (pastries) **before it opens**. Table 1 lists information it could use. Across 168 days of data, **the planned price and the new-price-period flag are identical ($r = 1.00$)**. Temperature has a correlation of -0.26 with pastry sales and the weekend flag 0.50 .

Table 1. Candidate information.

#	Information
1	Tomorrow’s temperature forecast (°C)
2	Tomorrow’s wind forecast (km/h)
3	Whether tomorrow is a weekend (yes or no)
4	Whether a beach event is on the calendar tomorrow (yes or no)
5	Tomorrow’s planned pastry price (€)
6	A “new price period” flag: 0 before 30 June, 1 from 30 June (the day the price rose)
7	Tomorrow’s gelato sales (cups), counted at closing time

1a Choose **three** inputs for the model. For each, say whether it should push pastry sales **up or down**, and give a one-line reason. 3 pts

Input (name or number)	Up or down?	Reason (one line)

1b One candidate in Table 1 is **unsuitable** for this forecast. Which one, and what is the problem? 2 pts

1c Should the planned price **and** the new-price-period flag both go into the same model? Explain, and say which one you would keep. 2 pts

1d How would you check the model’s prediction errors on days that were **not** used to fit it? 1 pt

SECTION 2 · Read the LINEST output

12 points · about 10 minutes

A model of daily pastry sales (pastries) was fitted to 168 days with $\text{=LINEST}(y, X, \text{TRUE}, \text{TRUE})$. The three X columns were entered in this order: X_1 = **temperature (°C)**, X_2 = **wind (km/h)**, X_3 = **weekend (1 = Saturday or Sunday, 0 = weekday)**. Use Table 2 to read Table 3.

Table 2. How LINEST lays out its results.

Col A (X_3)	Col B (X_2)	Col C (X_1)	Col D (Intercept)	Statistic description
Slope m_3	Slope m_2	Slope m_1	Intercept (b)	Coefficients for: $y = m_1X_1 + m_2X_2 + m_3X_3 + b$
SE – m_3	SE – m_2	SE – m_1	SE – Intercept	Standard error for each coefficient
R^2	SE – Y	Blank	Blank	Quality of the model fit
F-statistic	df	Blank	Blank	Analysis of variance metrics
SS – Reg	SS – Resid	Blank	Blank	Sum of squares metrics

Table 3. LINEST output for the model (shaded cells are blank).

Col A	Col B	Col C	Col D
11.53	–0.18	–0.81	84.76
1.53	0.10	0.20	4.98
0.33	8.95		
26.34	164		
6329.40	13133.74		

2a Write the fitted equation, using the numbers in Table 3 and the variable names.

2 pts

2b Each statement below misreads the output. Rewrite each one correctly, keeping the units and conditions.

3 pts

(i)	"The negative temperature coefficient means pastries do not sell on warm days."
(ii)	"Wind has a coefficient of only –0.18, so wind does not matter at all."
(iii)	" $R^2 = 0.33$ means the model is wrong 67% of the time."

2c (a) By how much does predicted sales change on a day that is **5 °C warmer**, everything else the same? (b) By how much does it change if the wind is **10 km/h stronger**? (c) The temperature coefficient is negative. Give one possible real-world reason.

3 pts

2d Tomorrow is forecast to be 20 °C with a 12 km/h wind on a weekend. (a) Predict pastry sales. (b) Actual sales turn out to be **98 pastries**: calculate the residual. (c) SE – Y is the typical size of a forecast error. Is your residual smaller or larger than SE – Y, and what does that say about this forecast?

4 pts

SECTION 3 · Errors and outliers

8 points · about 7 minutes

3a · Errors. Sales and errors are in pastries. These are constructed exercise numbers, not workbook output. **Residual = actual – predicted.** **MAE = (sum of the absolute residuals) ÷ n.**

Day	Actual	Predicted	Residual
Mon	62	66	
Sat		74	+10
Wed	58		-2
Fri	65	69	

3a-i Fill the blanks in the table.

1 pt

3a-ii Calculate the MAE of these predictions.

1 pt

3a-iii Model Z has residuals of -3, -3, -3, -3 on the same four days. Which model has the lower MAE? What do the identical signs tell you about Model Z?

2 pts

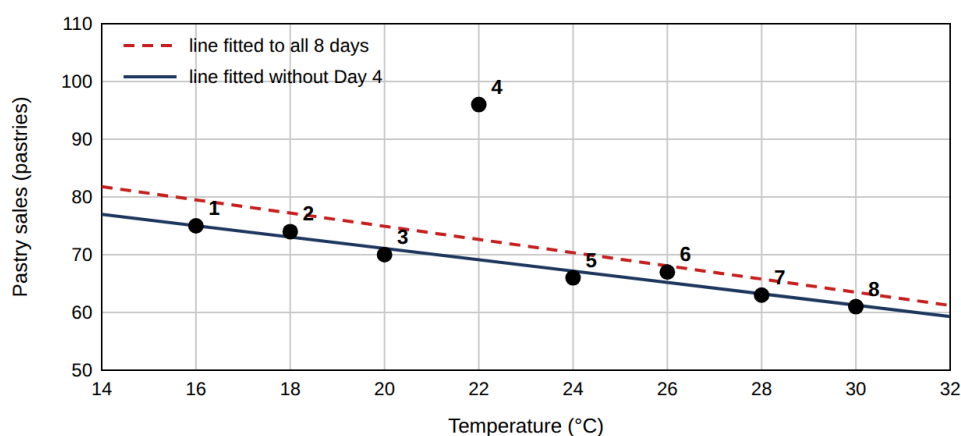
3b · Outliers. Table 4 and Figure 1 show pastry sales on 8 days (constructed example). Table 5 gives the straight-line results with and without Day 4.

Day	1	2	3	4	5	6	7	8
Temperature (°C)	16	18	20	22	24	26	28	30
Sales (pastries)	75	74	70	96	66	67	63	61

Table 5. Results for the straight line (sales against temperature).

	All 8 days	Without Day 4
Slope (pastries per °C)	-1.14	-0.98
R ²	0.26	0.96
SE – Y (pastries)	10.28	1.17

Figure 1. The 8 days, with the line fitted to all of them (dashed) and the line fitted without Day 4 (solid).



3b-i Which day does not follow the pattern of the others?

1 pt

3b-ii Compare the results with and without that day. What impact does it have on the slope, and what impact on R²?

2 pts

3b-iii Before deleting it, what would you do?

1 pt

SECTION 4 · Did the price change sales?

9 points · about 10 minutes

On 30 June the café raised the pastry price from **€2.50** to **€2.75**. Table 6 shows pastry sales (pastries) on 5 weekdays **before** (Group 1) and 5 weekdays **after** (Group 2). Both samples are weekdays with similar weather.

Formulas (as given in class):

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

A negative t only means that Group 2 has the larger mean.

Table 6. Worksheet (fill in the blank cells).

Day	Before (x ₁)	x ₁ - $\bar{x}_1$	(x ₁ - $\bar{x}_1$) ²	After (x ₂)	x ₂ - $\bar{x}_2$	(x ₂ - $\bar{x}_2$) ²
1	66			63		
2	70			67		
3	62			59		
4	68			65		
5	64			61		
Sum						
Mean $\bar{x}$						
s						

4a Complete Table 6: the two means, the two sums of squared deviations, and s for each group (2 decimals). **3 pts**

4b Calculate s_p and t (2 decimals). **2 pts**

Sheets gives **=T.TEST(B2:B6, C2:C6, 2, 2) = 0.172** (two-tailed, equal variances).

4c (i) State the null hypothesis in words. (ii) Is the difference statistically significant at 5%? Explain using the p-value. (iii) Does this result prove the price rise had **no** effect? Explain. **3 pts**

4d Name one way to make a price effect easier to detect next time. **1 pt**

SECTION 5 · Summary for the owner

3 points · about 3 minutes

5 Write 3–4 sentences to the café owner, who has no statistics background. Include: (i) which inputs you recommend for forecasting pastry sales and the main finding from the LINEST output; (ii) what the price test showed; (iii) one caution. **3 pts**

— End of practice —